space dimensions, it would be an inverse fifth-power law (double the separation, force drops by a factor of 32); and so on for ever higher-dimensional universes.

You might think that the success of the inverse square version of Newton's law in explaining a wealth of data—from the motion of planets to the paths of comets—confirms that we live in a universe with precisely three space dimensions. But that conclusion would be hasty. We know that the inverse square law works on astronomical scales,[6] and we know that it works on terrestrial scales, and that jibes well with the fact that on such scales we see three space dimensions. But do we know that it works on smaller scales? How far down into the microcosmos has gravity's inverse square law been tested? As it turns out, experimenters have confirmed it down to only about a tenth of a millimeter; if two objects are brought to within a separation of a tenth of a millimeter, the data verify that the strength of their gravitational attraction follows the predictions of the inverse square law. But so far, it has proven a significant technical challenge to test the inverse square law on shorter scales (quantum effects and the weakness of gravity complicate the experiments). This is a critical issue, because deviations from the inverse square law would be a convincing signal of extra dimensions.

To see this explicitly, let's work with a lower-dimensional toy example that we can easily draw and analyze. Imagine we lived in a universe with one space dimension—or so we thought, because only one space dimension was visible and, moreover, centuries of experiments had shown that the force of gravity does not vary with the separation between objects. But also imagine that in all those years experimenters had been able only to test the law of gravity down to distances of about a tenth of a millimeter. For distances shorter than that, no one had any data. Now, imagine further that, unbeknownst to everyone but a handful of fringe theoretical physicists, the universe actually had a second, curled-up space dimension making its shape like the surface of Philippe Petit's tightrope, as in Figure 12.5. How would this affect future, more refined gravitational tests? We can deduce the answer by looking at Figure 13.6. As two tiny objects are brought close enough together—much closer than the circumference of the curled-up dimension—the two-dimensional character of space would become apparent immediately, because on those scales gravitational field lines *would* have room to